Execution Screenshots: End-to-End Recommendation Data Pipeline

Roll No: 2025AG05312 | AIML ZG529 Data Management for Machine Learning, Assignment 2

GitHub repository: https://github.com/jayatimishra/dm4ml-recsys-2025AG05312

Every terminal panel below is the captured output of the command shown, produced by `tools/capture_execution.py`, which deletes all generated data and runs each stage in order from an empty data folder. Full untrimmed reference output: `docs/execution/captures.json`. Reference run summary:

Step	Command	Exit	Seconds
Environment	`python --version && pip show pandas mlflow prefect dvc pandera 2>/dev/null | grep -E '^(Name|Ve...`	0	0.5
Source systems (synthetic data)	`python -m src.ingestion.generate_sources --end-date 2026-11-01`	0	1.6
Stage 1: Data ingestion	`python -m src.ingestion.ingest --mode incremental && find data/raw -type f | sort`	0	1.8
Stage 2: Data validation	`python -m src.validation.validate && echo && head -5 data/errors/interactions_rejected.csv | cu...`	0	0.9
Stage 3: Data preparation	`python -m src.preparation.prepare && ls reports/eda`	0	1.8
Stage 4: Data transformation	`python -m src.transformation.transform && python -c "import sqlite3; c=sqlite3.connect('data/wa...`	0	4.4
Stage 5: Feature management	`python -m src.features.feature_store && python -c "import sys; sys.path.insert(0,'.'); from src...`	0	1.5
Stage 6: Pre-training validation	`python -m src.validation.pre_training_checks`	0	0.8
Stage 7: Model training (MLflow)	`python -m src.training.train`	0	11.5
Stage 8: Model evaluation	`python -m src.evaluation.evaluate && tail -n +3 reports/model/results_table.md`	0	2.9
Stage 9: Prediction service	`python -m src.serving.serve --batch && { python -m src.serving.serve --serve > /tmp/serve_captu...`	0	7.8
Orchestration: full DAG run	`python dags/run_pipeline.py 2>&1 | grep orchestrator`	0	19.6
Versioning: DVC pipeline	`dvc dag && sed -n '/dvc params diff/,/dvc metrics diff/p' reports/versioning/dvc_evidence.txt |...`	0	0.8

Captured 2026-11-06. All steps exited with code 0.

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Setup

Environment and Git history (tags data-v1, data-v2)

Environment | Roll No: 2025AG05312 | 2026-11-06 03:56:48

```
[2025AG05312] dm4ml-recsys $ python --version && pip show pandas mlflow prefect dvc pandera 2>/dev/null | grep -E '^(Name|Version)' | paste - - && git l
og --oneline --decorate -3
Python 3.13.18
Name: pandas    Version: 3.0.5
Name: mlflow    Version: 3.17.2
Name: prefect   Version: 3.9.1
Name: dvc       Version: 3.67.1
Name: pandera   Version: 0.35.0
8f84465 (HEAD -> main) Final submission: docs, report, evaluation, watermark tooling
8e8b9ae (tag: data-v2) Data v2: full history to 2026-10-31 via incremental ingestion
412aab6 (tag: data-v1) Data v1: events up to 2026-10-17 (pipeline run 1)
[exit code 0]
```

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1. Data ingestion

Synthetic source systems generated (documented, seed 529)

Source systems (synthetic data) | Roll No: 2025AG05312 | 2026-11-06 03:56:50

```
[2025AG05312] dm4ml-recsys $ python -m src.ingestion.generate_sources --end-date 2026-11-01
2026-11-06 03:56:50,247 | ingestion.generate | INFO | Generated sources for 2025AG05312: events=77217 users=1505 products=604 (end_date=2026-11-01)
[exit code 0]
```

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Ingestion of CSV exports + paginated REST API into the partitioned raw zone

Stage 1: Data ingestion | Roll No: 2025AG05312 | 2026-11-06 03:56:52

```
[2025AG05312] dm4ml-recsys $ python -m src.ingestion.ingest --mode incremental && find data/raw -type f | sort
2026-11-06 03:56:51,019 | ingestion.ingest | INFO | clickstream: ingested 77217 rows (incremental), hwm None -> e0076302
2026-11-06 03:56:51,028 | ingestion.ingest | INFO | crm users: ingested 1505 rows (full snapshot)
2026-11-06 03:56:51,044 | ingestion.ingest | WARNING | API http://127.0.0.1:8529/api/v1/products?page=3&page_size=100 returned 503, retry 1/3 in 0.5s
2026-11-06 03:56:52,064 | ingestion.ingest | INFO | catalog api: ingested 604 records over 7 pages
data/raw/_state/ingestion_state.json
data/raw/catalog_api/products/ingest_date=2026-11-06/batch_20261106035650.json
data/raw/catalog_api/products/ingest_date=2026-11-06/batch_20261106035650.manifest.json
data/raw/clickstream/interactions/ingest_date=2026-11-06/batch_20261106035650.csv
data/raw/clickstream/interactions/ingest_date=2026-11-06/batch_20261106035650.manifest.json
data/raw/crm/users/ingest_date=2026-11-06/batch_20261106035650.csv
data/raw/crm/users/ingest_date=2026-11-06/batch_20261106035650.manifest.json
[exit code 0]
```

Roll No: 2025AG05312 | captured 2026-11-06 03:56:52

Roll No: 2025AG05312

The WARNING line is the deliberate HTTP 503 from the mock API, recovered by retry with backoff.

2. Data validation

Validation run, quality gate and the error dataset (data/errors)

Stage 2: Data validation | Roll No: 2025AG05312 | 2026-11-06 03:56:53

```
[2025AG05312] dm4ml-recsys $ python -m src.validation.validate && echo && head -5 data/errors/interactions_rejected.csv | cut -c1-150
2026-11-06 03:56:52,689 | validation | INFO | users: in=1505 dup_dropped=5 rejected=0 passed=1500
2026-11-06 03:56:52,719 | validation | INFO | products: in=604 dup_dropped=4 rejected=0 passed=600
2026-11-06 03:56:52,974 | validation | INFO | interactions: in=77217 dup_dropped=915 rejected=379 passed=75923
2026-11-06 03:56:53,028 | validation | INFO | Validation done: DQ score=0.9980 gate=PASSED

# Roll No: 2025AG05312 | AIML ZG529 Data Management for Machine Learning
event_id,user_id,product_id,event_type,rating,event_timestamp,session_id,device,failed_checks
e0000095,,1467,view,,2026-05-14 14:06:17,s830_9,mobile,INT_002;INT_010
e0000207,,1392,view,,2026-05-18 00:35:32,s36_13,tablet,INT_002;INT_010
e0000218,,1145,view,,2026-05-18 11:01:33,s633_13,desktop,INT_002;INT_010
[exit code 0]
```

Roll No: 2025AG05312 | captured 2026-11-06 03:56:53

Roll No: 2025AG05312

Validation report (reports/validation/validation_report.html)

Data Validation Report

Roll No: 2025AG05312 | run at 2026-11-06T01:06:53.974335 | overall gate: **PASSED**

Entity summary

entity	rows_in	duplicates_dropped	rows_rejected	rows_passed
users	1505	5	0	1500
products	604	4	0	600
interactions	77217	915	379	75923

Quality gate (dataset-level)

entity	check	result	detail
users	schema_columns_present	PASS	missing=[] unexpected=[]
users	min_row_count	PASS	rows=1505 min=100
users	null_rate[user_id]<=0.0	PASS	null_rate=0.0000
users	critical_reject_rate<=5%	PASS	reject_rate=0.0000
products	schema_columns_present	PASS	missing=[] unexpected=[]
products	min_row_count	PASS	rows=604 min=100
products	null_rate[product_id]<=0.0	PASS	null_rate=0.0000
products	critical_reject_rate<=5%	PASS	reject_rate=0.0000
interactions	schema_columns_present	PASS	missing=[] unexpected=[]
interactions	min_row_count	PASS	rows=77217 min=1000
interactions	null_rate[user_id]<=0.01	PASS	null_rate=0.0020
interactions	null_rate[product_id]<=0.01	PASS	null_rate=0.0000
interactions	freshness<=7d	PASS	latest=2026-10-31 23:57:56 staleness_days=5
interactions	critical_reject_rate<=5%	PASS	reject_rate=0.0050

Row-level checks

entity	id	column	dimension	severity	description	checked	failed	pass rate
users	USR_UNQ	user_id	uniqueness	critical	user_id is unique (duplicates dropped, first kept)	1505	5	0.9967
users	USR_001	user_id	completeness	critical	user_id is not null	1500	0	1.0000
users	USR_002	signup_date	validity	critical	signup_date is a valid date	1500	0	1.0000
users	USR_003	age	completeness	warning	age is not null	1500	20	0.9867

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3. Data preparation

Cleaning, standardisation and EDA

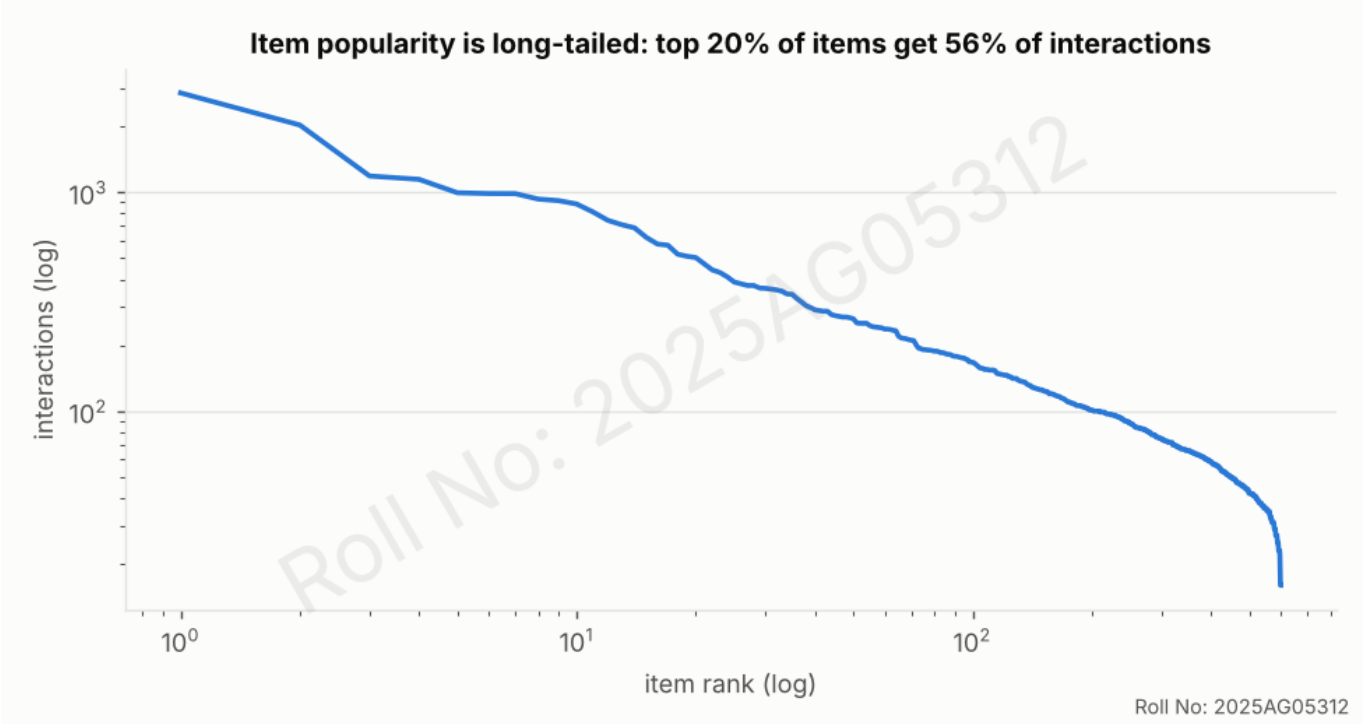
Stage 3: Data preparation | Roll No: 2025AG05312 | 2026-11-06 03:56:54

```
[2025AG05312] dm4ml-recsys $ python -m src.preparation.prepare && ls reports/eda
2026-11-06 03:56:54,830 | preparation | INFO | Prepared users=1500 products=600 interactions=75923; repairs={'users.age_imputed_with_median': 35, 'users.gender_set_unknown': 5, 'users.city_filled_unknown': 45, 'products.category_casing_fixed': 3, 'products.category_inferred_from_title': 12, 'products.price_text_parsed': 6, 'products.price_imputed_category_median': 15, 'products.price_winsorised_p99_5': 3, 'interactions.event_type_normalised': 305, 'interactions.rating_invalid_set_null': 53}
01_daily_events.png
02_funnel.png
03_item_long_tail.png
04_user_activity.png
05_ratings.png
06_category_share.png
07_hour_of_day.png
eda_summary.json
[exit code 0]
```

Roll No: 2025AG05312 | captured 2026-11-06 03:56:54

Roll No: 2025AG05312

EDA: item long tail



4. Data transformation

Star schema load, point-in-time features, time-based splits, derived datasets

Stage 4: Data transformation | Roll No: 2025AG05312 | 2026-11-06 03:56:59

```
[2025AG05312] dm4ml-recsys $ python -m src.transformation.transform && python -c "import sqlite3; c=sqlite3.connect('data/warehouse/recsys_warehouse.db'); [print(f'{t:22s}', c.execute(f'select count(*) from...
2026-11-06 03:56:56,173 | transformation | INFO | Star schema loaded: {'dim_user': 1500, 'dim_product': 600, 'dim_date': 180, 'fact_interactions': 75923}
2026-11-06 03:56:56,778 | transformation | INFO | Features materialised as_of=2026-10-04 00:00:00: {'feat_user': 1500, 'feat_item': 600, 'agg_user_item': 26433, 'feat_user_category': 9371}
2026-11-06 03:56:57,576 | transformation | INFO | Features materialised as_of=2026-10-18 00:00:00: {'feat_user': 1500, 'feat_item': 600, 'agg_user_item': 33353, 'feat_user_category': 10891}
2026-11-06 03:56:58,549 | transformation | INFO | Features materialised as_of=2026-11-01 00:00:00: {'feat_user': 1500, 'feat_item': 600, 'agg_user_item': 40220, 'feat_user_category': 11883}
2026-11-06 03:56:59,172 | transformation | INFO | Splits exported: {'train': 61383, 'valid': 13680, 'test': 14540}
dim_user               1500
dim_product            600
dim_date               180
fact_interactions      75923
feat_user              4500
feat_item              1800
agg_user_item          100006
data/processed/derived:
item_popularity.parquet
matrix_item_index.parquet
matrix_user_index.parquet
user_activity_summary.parquet
user_item_interactions_long.parquet
user_item_matrix_train.npz

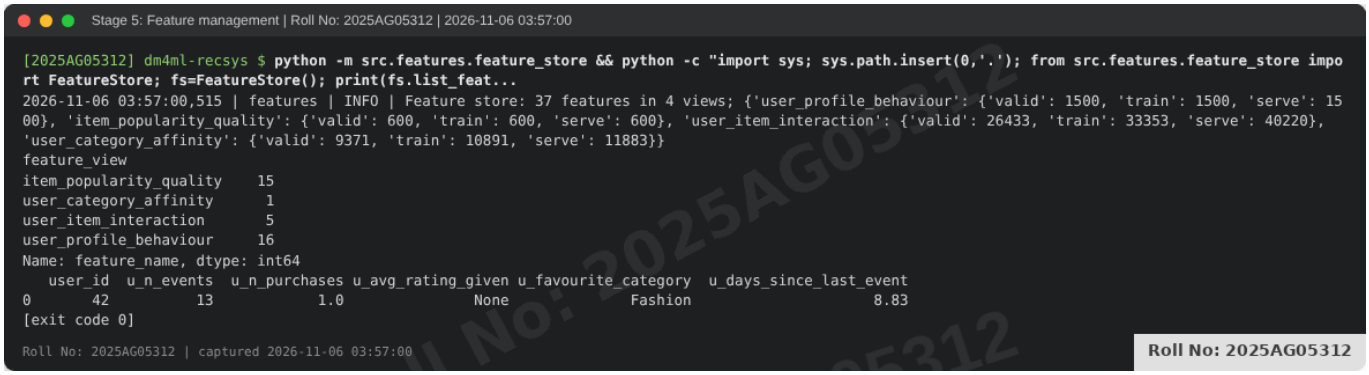
data/processed/splits:
test_interactions.parquet
train_interactions.parquet
valid_interactions.parquet
[exit code 0]
```

Roll No: 2025AG05312 | captured 2026-11-06 03:56:59

Roll No: 2025AG05312

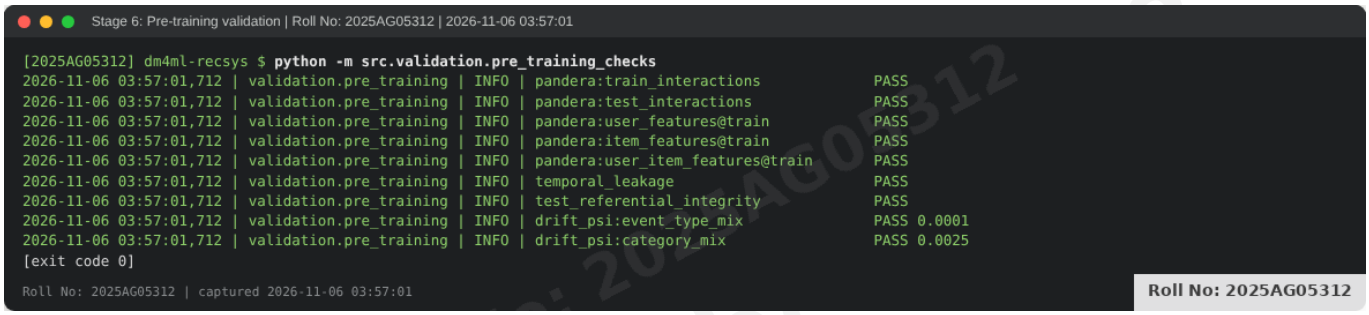
5. Feature management

Feature registry, offline/online materialisation and an online lookup



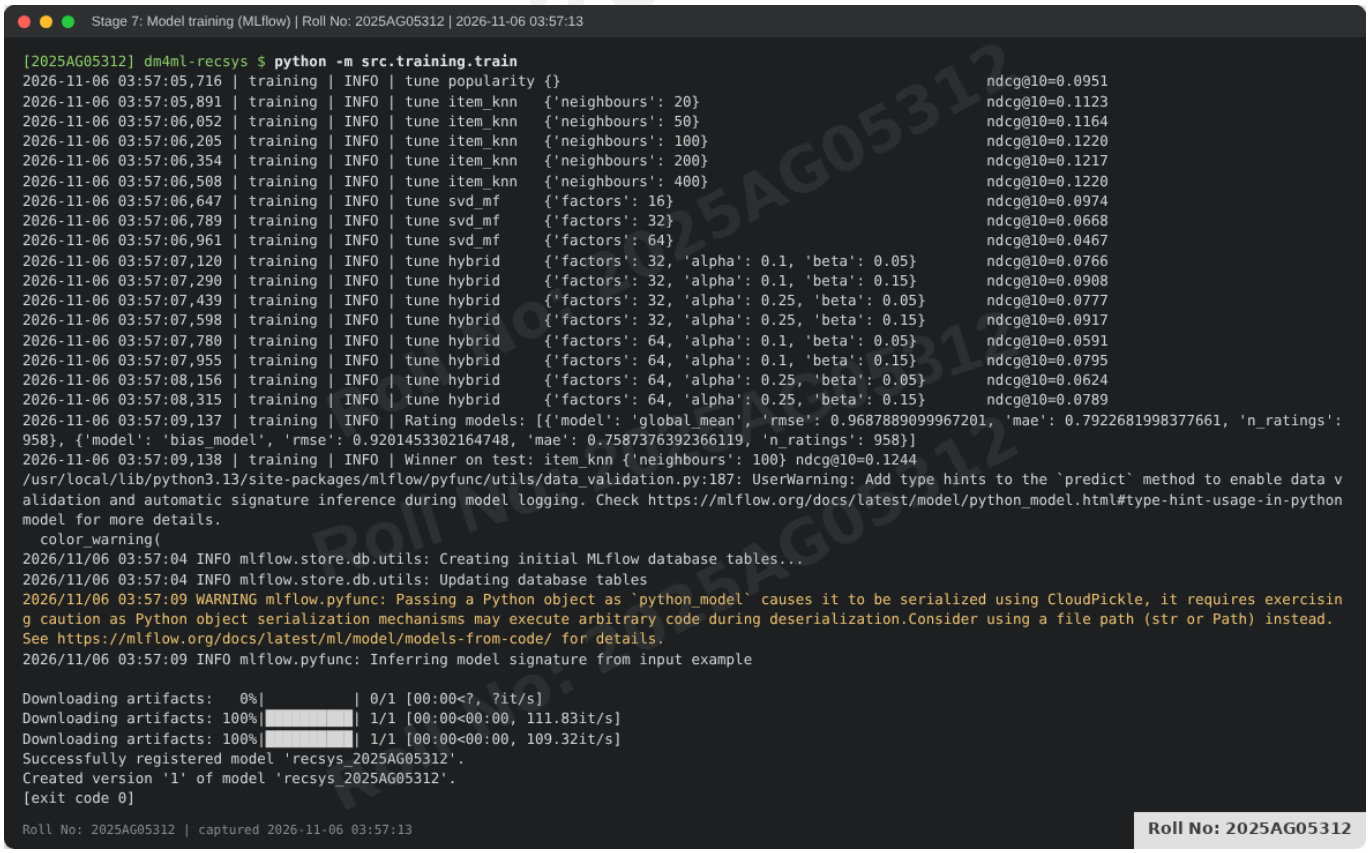
6. Pre-training validation

Pandera schemas, temporal leakage check, PSI drift

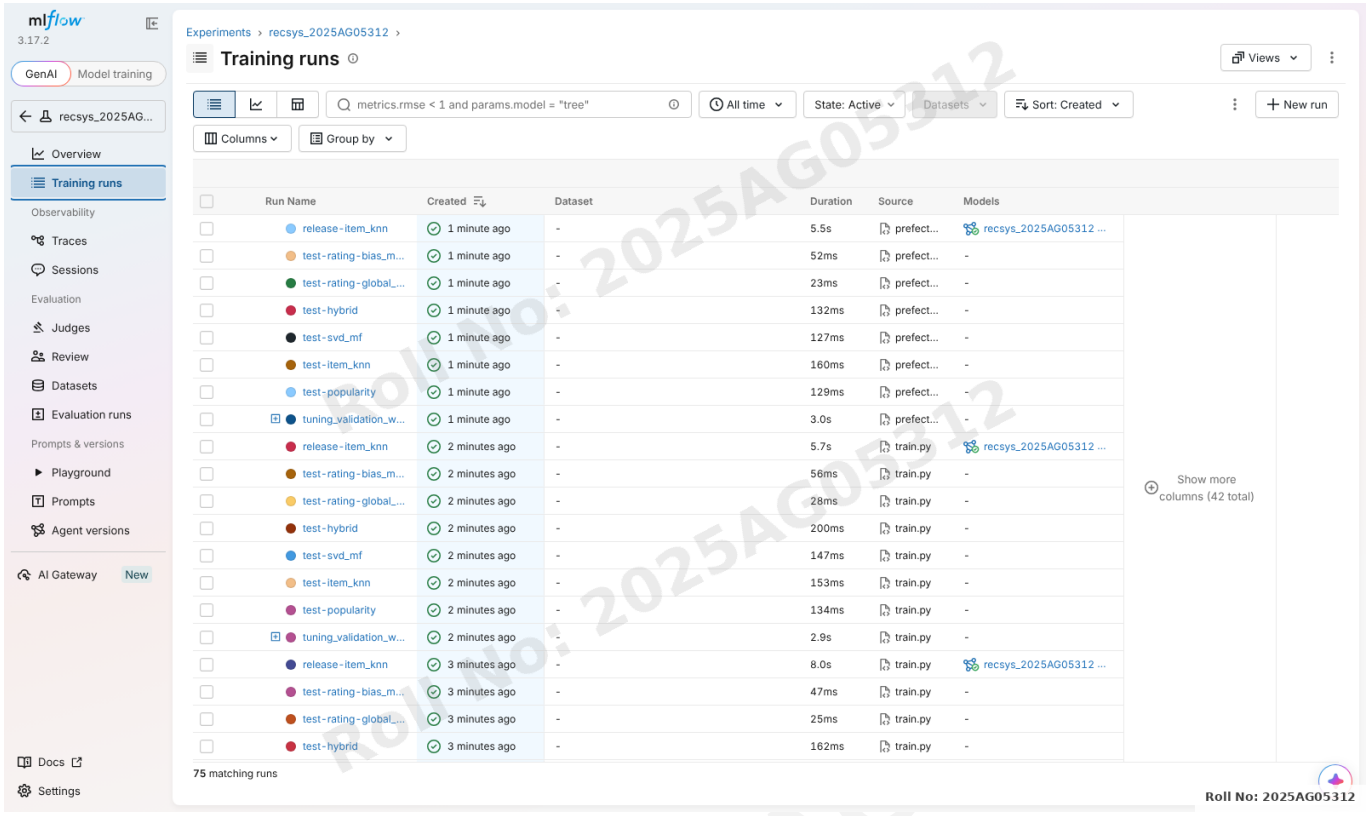


7. Model training and experiment tracking

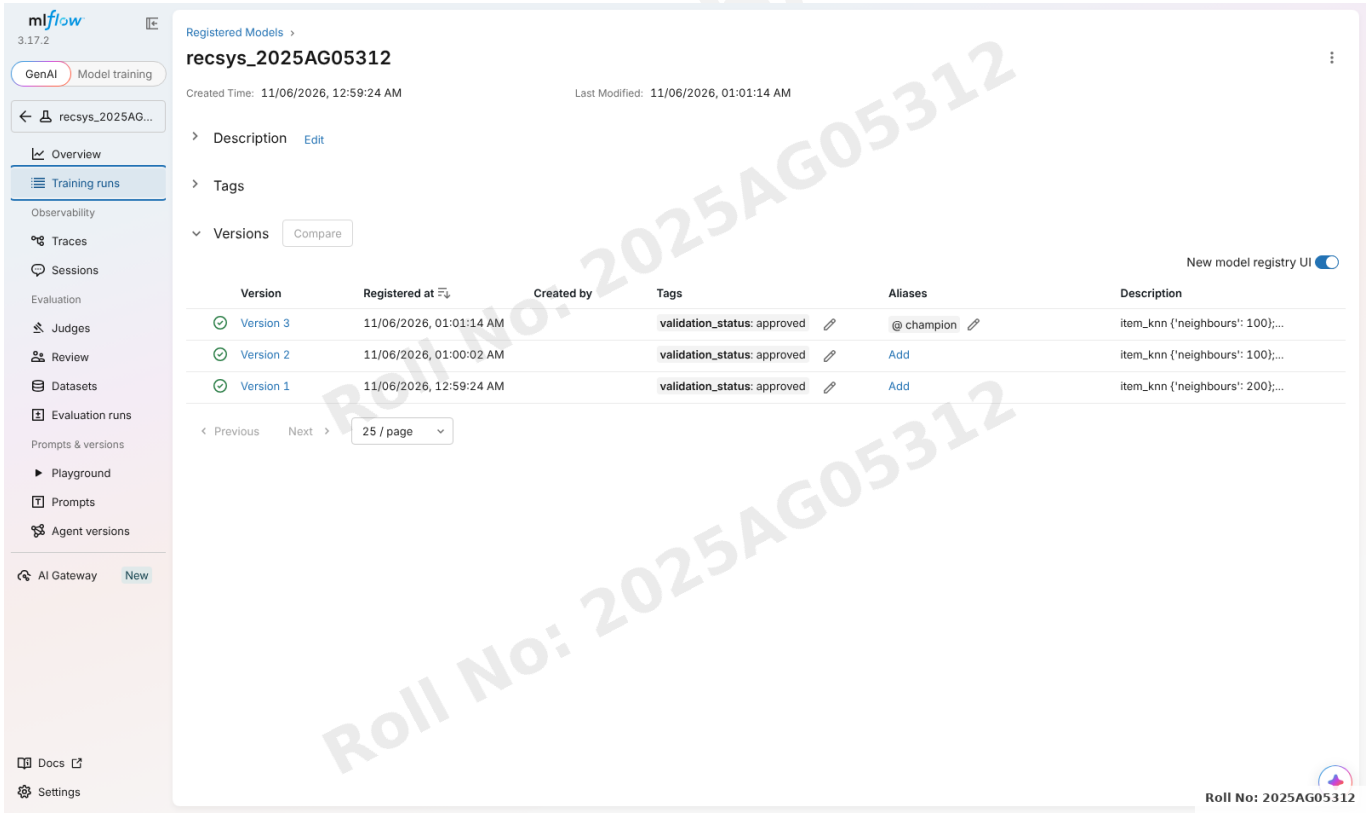
Tuning on validation window, one-shot test, release + MLflow registry



MLflow: experiment runs



MLflow: model registry with alias 'champion'



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8. Model evaluation

Results table comparing all models

Stage 8: Model evaluation | Roll No: 2025AG05312 | 2026-11-06 03:57:16

```
[2025AG05312] dm4ml-recsys $ python -m src.evaluation.evaluate && tail -n +3 reports/model/results_table.md
2026-11-06 03:57:15,904 | evaluation | INFO | Evaluation table written; winner=item_knn ndcg@10=0.1244 vs baseline 0.0851

Evaluated on 921 users with at least one new add-to-cart or purchase in the 14-day test window. Selected model: **Item-KNN CF**.

| Model | Params | precision@10 | recall@10 | ndcg@10 | map@10 | hit_rate@10 | coverage | diversity | novelty | NDCG lift |
|---|---|---|---|---|---|---|---|---|---|---|
| Popularity (baseline) | `{}` | 0.0312 | 0.1312 | 0.0851 | 0.0526 | 0.2628 | 0.0683 | 0.8963 | 6.69 | +0.0% |
| **Item-KNN CF** | `{"neighbours": 100}` | 0.0451 | 0.1850 | 0.1244 | 0.0789 | 0.3529 | 0.4550 | 0.7191 | 7.53 | +46.1% |
| SVD matrix factorisation | `{"factors": 16}` | 0.0367 | 0.1504 | 0.0990 | 0.0605 | 0.2964 | 0.3817 | 0.6526 | 8.01 | +16.3% |
| Hybrid (SVD + content + popularity) | `{"factors": 32, "alpha": 0.25, "beta": 0.15}` | 0.0359 | 0.1505 | 0.1007 | 0.0624 | 0.2899 | 0.7100 | 0.3407 | 8.46 | +18.3% |

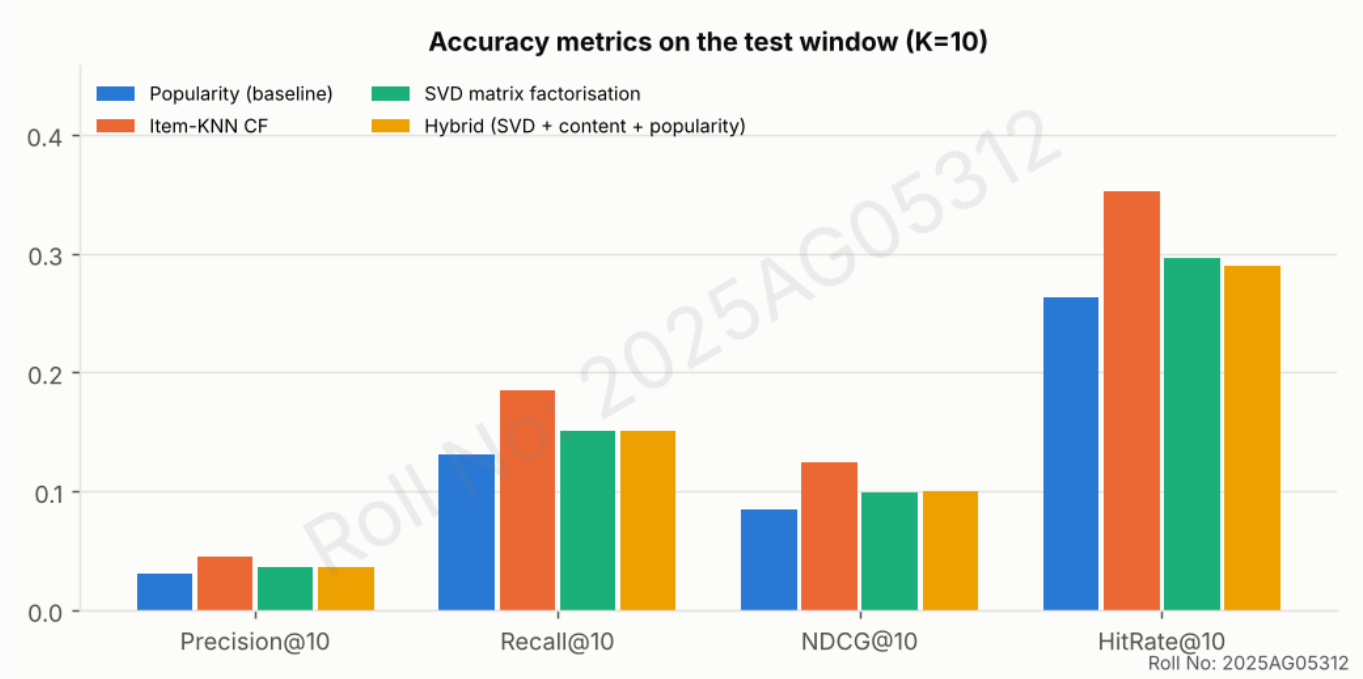
## Explicit rating prediction (supplementary)

| Model | RMSE | MAE | Test ratings |
|---|---|---|---|
| global_mean | 0.9688 | 0.7923 | 958 |
| bias_model | 0.9201 | 0.7587 | 958 |
[exit code 0]
```

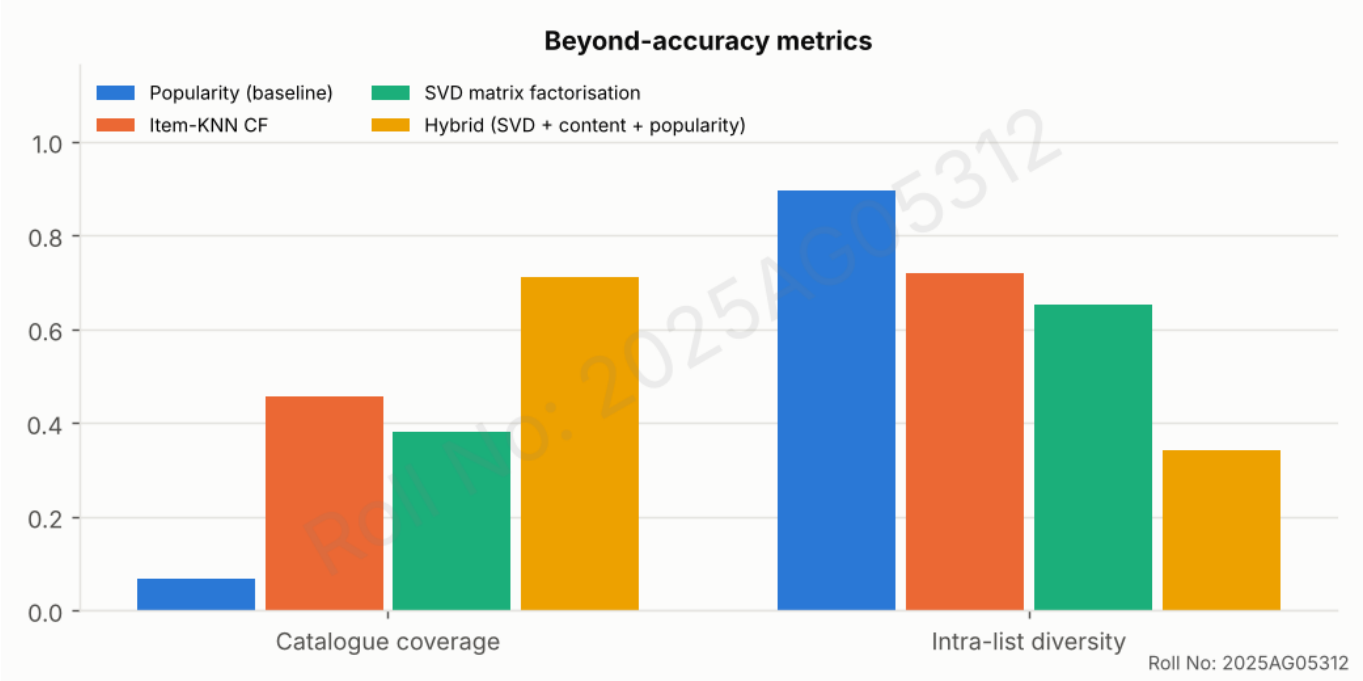
Roll No: 2025AG05312 | captured 2026-11-06 03:57:16

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Accuracy metrics



Beyond-accuracy metrics



9. Prediction service

Batch recommendations and HTTP /recommend endpoint

Stage 9: Prediction service | Roll No: 2025AG05312 | 2026-11-06 03:57:24

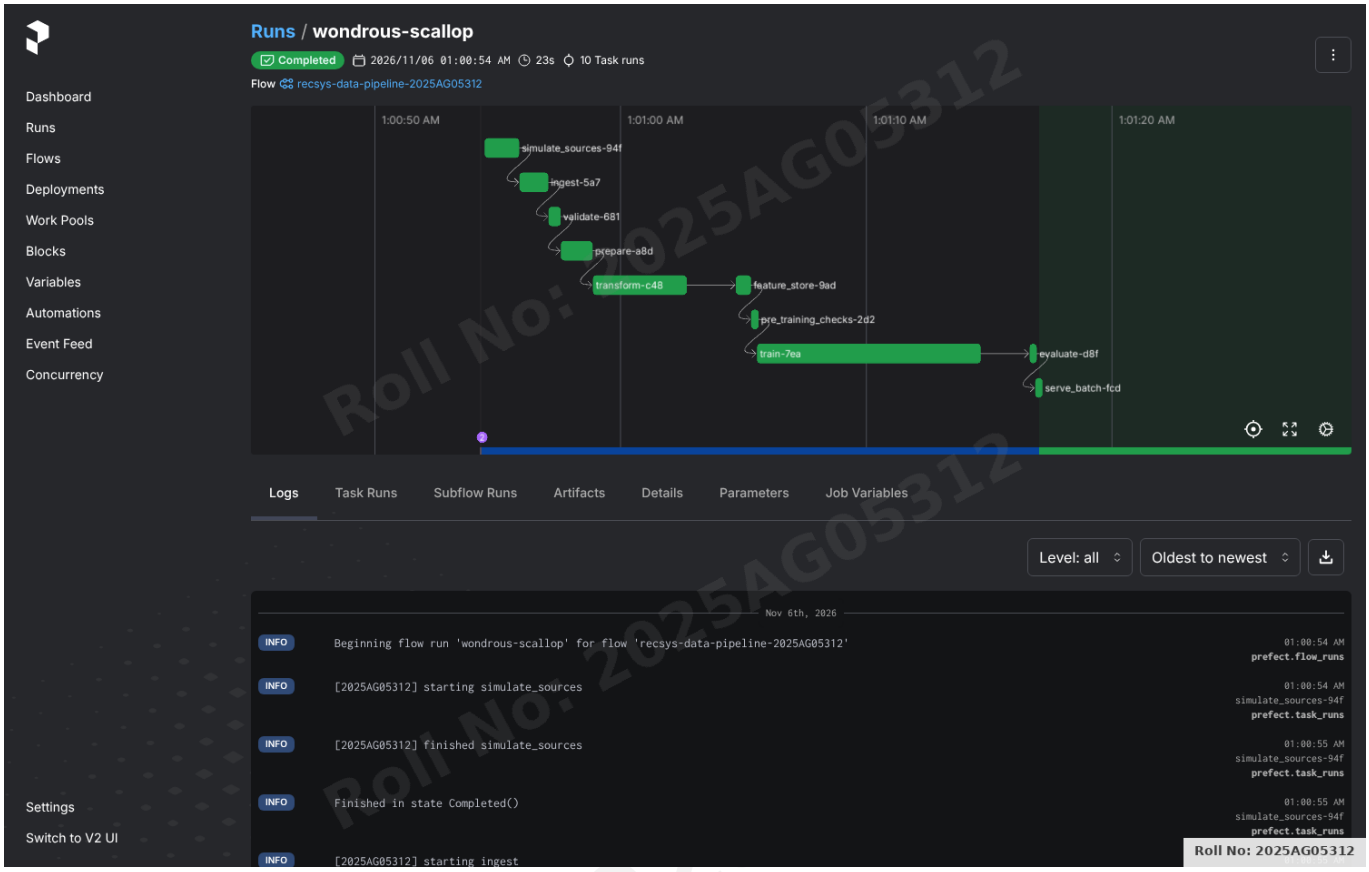
```
[2025AG05312] dm4ml-recsys $ python -m src.serving.serve --batch && { python -m src.serving.serve --serve > /tmp/serve_capture.log 2>&1 & SRV=$!; } ; fo
r i in $(seq 1 40); do sleep 1; curl -s 'http://127.0....
2026-11-06 03:57:19,715 | serving | INFO | Batch recommendations for [7, 42, 314, 1001, 99999] written (model v1); user 99999 is a cold-start test
{
    "served_by": "2025AG05312",
    "recommendations": [
        {
            "user_id": 42,
            "rank": 1,
            "product_id": 1327,
            "score": 1.14614,
            "category": "Automotive",
            "brand": "BrandQ",
            "price": 3210.68,
            "i_avg_rating": 3.379,
            "in_stock": 1,
            "model_version": 1
        },
        {
            "user_id": 42,
            "rank": 2,
            "product_id": 1160,
            "score": 0.95491,
            "category": "Sports",
... (7 lines omitted, full output in docs/execution/captures.json) ...
            "user_id": 42,
            "rank": 3,
            "product_id": 1472,
            "score": 0.87505,
            "category": "Fashion",
            "brand": "BrandL",
            "price": 1202.66,
            "i_avg_rating": 4.074,
            "in_stock": 1,
            "model_version": 1
        },
        {
/usr/local/lib/python3.13/site-packages/mlflow/pyfunc/utils/data_validation.py:187: UserWarning: Add type hints to the `predict` method to enable data v
alidation and automatic signature inference during model logging. Check https://mlflow.org/docs/latest/model/python_model.html#type-hint-usage-in-python
model for more details.
  color_warning(
[exit code 0]
```

Roll No: 2025AG05312 | captured 2026-11-06 03:57:24

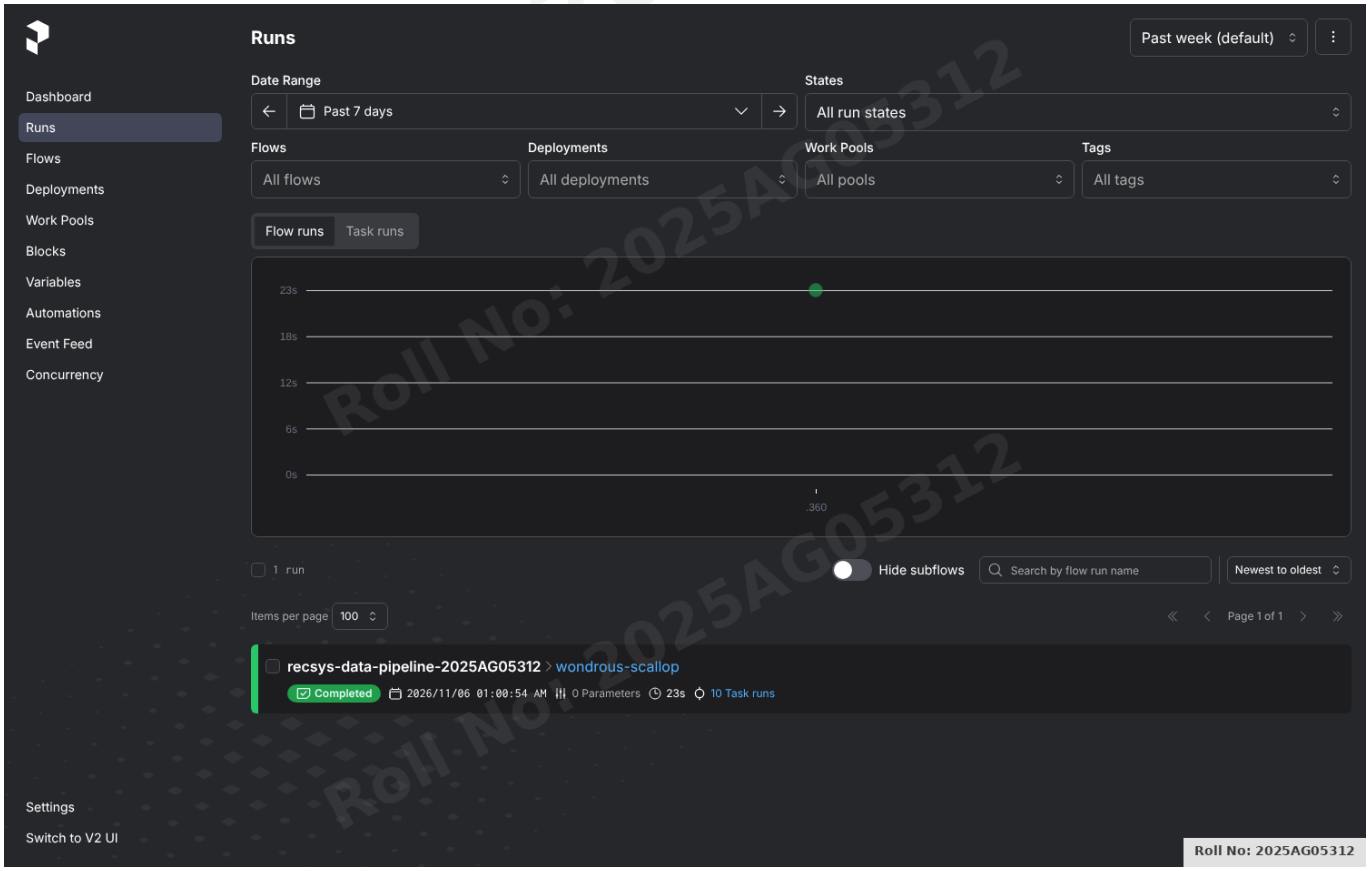
Roll No: 2025AG05312

10. Orchestration and reproducibility

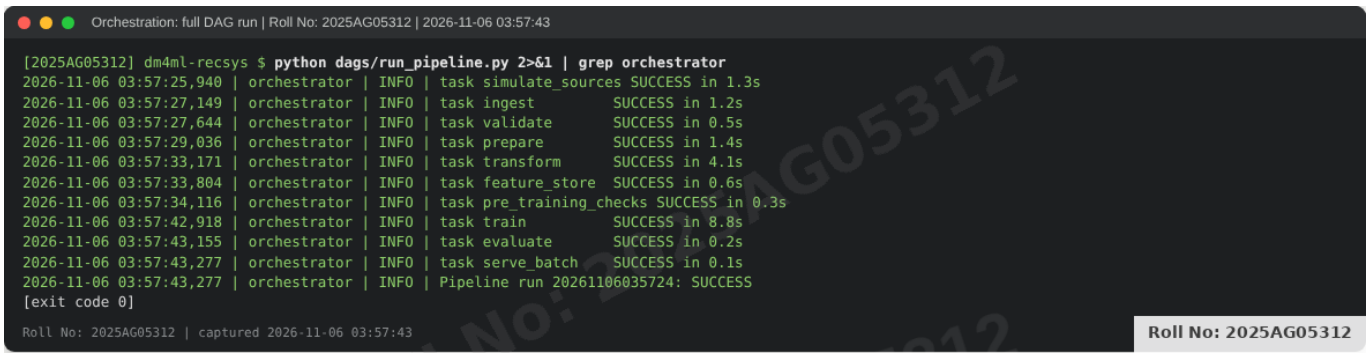
Prefect flow run: all tasks Completed



Prefect: run history



Full DAG run with the dependency-free runner



DVC pipeline DAG and data-version diff

